EXPERIMENT – 5

AIM :

To write a Python program to solve the Missionaries and Cannibals Problem by safely transporting all missionaries and cannibals across a river without violating the given constraints.

ALGORITHM:

1. Start.
2. Initialize the number of missionaries and cannibals on both riverbanks.
3. Place the boat on the starting bank.
4. Move one or two people across the river using the boat.
5. Ensure that cannibals never outnumber missionaries where missionaries are present.
6. Repeat the crossing process until everyone reaches the opposite bank.
7. Display the sequence of valid crossings.
8. Stop.

PROGRAM :

untitled (3.14.3)

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```
from collections import deque
def valid(m, c):
    if not (0 <= m <= 3 and 0 <= c <= 3):
        return False
    # Left bank
    if m > 0 and c > m:
        return False
    # Right bank
    mr, cr = 3 - m, 3 - c
    if mr > 0 and cr > mr:
        return False
    return True
def solve():
    start = (3, 3, 0)  # M, C, boat: 0=left, 1=right
    goal = (0, 0, 1)
    moves = [(1, 0), (2, 0), (0, 1),
             (0, 2), (1, 1)]
    queue = deque([(start, [start])])
    visited = {start}
    while queue:
        state, path = queue.popleft()

        if state == goal:
            return path

        m, c, boat = state

        for dm, dc in moves:
            if boat == 0:
                new_state = (m-dm, c-dc, 1)
            else:
                new_state = (m+dm, c+dc, 0)

            if (valid(new_state[0], new_state[1])
                    and new_state not in visited):
                visited.add(new_state)
                queue.append((new_state, path + [new_state]))

    return None

path = solve()

if path:
    for state in path:
        print("Missionaries:", state[0],
              "Cannibals:", state[1],
              "Boat:", "Right" if state[2] else "Left")
else:
    print("No solution")
```

OUTPUT:



RESULT:

Thus, the Python program to solve the Missionaries and Cannibals Problem was executed successfully, and a valid sequence of river crossings was obtained.